
PalmSource™ 99
Adding Communications to Your App/Serial
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Introduction

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- Designing your app to communicate
- Pitfalls of communications
- Being Palm™ device-friendly
- Being responsive
- Serial APIs

Designing Your App to Communicate

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- Bolting Comms onto an existing app
- Porting from another platform
- Instead, get the UI and the Comms to work hand in hand

Pitfalls of Communications

- Blocking and the perils therein
- Being responsive to the user
- Being responsive to the other side

What Is Blocking

- Long reads or writes without timeouts
- Stops the UI in its tracks
- Prevents the app from responding to errors
- The user can’t interact either

Palm™ Device-Friendly Communications

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- Keep the Zen of Palm uppermost
- Design around presenting the data
- Keep the communications “in the background”

Designing around the Data

- Presenting the data
- Manipulating the data
- Sending and receiving are usually secondary

Backgrounding the Communications

- Work with the event loop
- Be quick about it!
- Don’t wait
- Don’t make them wait

Work with the Event Loop

- Short timeouts
- Polling communications

Be Quick about It

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- Get what's available then get out
- Always check what's available and what can be sent
- Keep the incoming buffer as empty as you can
- Keep the outgoing buffer as full as you can

Example of Event Loop with Comms Embedded

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```
EventType   event;
do {
   EvtGetEvent(&event, 10);
   HandleEvent(&event);
   if (sendPending) {
      SendSome();
   }
   error = SrmReceiveCheck(port, &pending;
   if (error) {
      HandleSerialError(error);
   }
   if (pending == 0) {
      TimedRead();
   }
   else {
      ReadSome(pending);
   }
}
while (event.eType != appStopEvent);
```

Serial API Overview

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- Not exhaustive
- Focused on performance
- Focused on responsiveness

Serial API Overview (Cont.)

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- SrmReceive
  - Use the timeout to advantage
  - Check the number of bytes received—may be less than you're looking for
  - Check the error result!

Serial API Overview (Cont.)

- SrmSend
  - Check the number of bytes sent—may be less than you gave
  - Check the error result!
  - Don’t send too much at a time

Serial API Overview (Cont.)

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- SrmReceiveWait
  - Avoid this function; exception: use a short timeout
  - Check the error result!

Serial API Overview (Cont.)

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- SrmReceiveCheck
  - Check the error result! (Do you get the feeling there's a message here?)
  - Use this as often as possible

Serial API Overview (Cont.)

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- SrmReceiveFlush
  - Be careful with the timeout, it’s a bit non-obvious

Serial API Overview (Cont.)

- SrmSetWakeupHandler
  - The handler is called at interrupt time
  - Use when fast responses are needed

Serial API Overview (Cont.)

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- SrmPrimeWakeupHandler
  - Prepares the wakeup handler

Serial API Overview (Cont.)

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- SrmReceiveWindowOpen
- SrmReceiveWindowClose

- Avoid these unless you really know what you're doing

Keeping Things Responsive

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- Keep the event loop running!
- Use short time outs
- Only read what you know is available

Q&A


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